Team name	[Your Team Name]
Project title	AcademiaFlow: Unified Student Lifecycle Companion for Course Planning, Accountable Groupwork, Verified Internships, and Thesis SLA Tracking
Selected case	Case 1: Smarter Campus and Learning
Members	[Member A (leader), Member B, Member C, Member D]

Abstract

1. Problem. Across their university lifecycle, undergraduates face four connected bottlenecks caused by rigid legacy portals and fragmented chat groups: (a) during course registration (KRS/FRS), students lack weekly syllabus previews and get separated from their study circle during parallel class selection; (b) in group projects (Tugas Besar), lecturers cannot see individual progress while students struggle with task handovers and free-riders; (c) in semesters 5–7, scholarship and MoU-internship posters get buried in WhatsApp groups, causing slow manual consultations over credit (SKS) conversion and alumni contacts; and (d) during thesis (Tugas Akhir) supervision, extended advisor inactivity stalls graduation without coordinator alerts.

2. Intended Users. First- through final-year undergraduate students, course lecturers supervising team assignments, and program coordinators (Kaprodi/Dosen Wali) overseeing credit conversions and thesis advisor workloads.

3. Supporting Observations or Evidence. From multi-campus workflow observations and a preliminary survey of [N, e.g., 45] undergraduates [1], [X, e.g., 37] experienced cohort separation or schedule clashes during KRS registration, and [*Y*, e.g., 39] faced unequal group workloads without lecturer visibility. Moreover, over 75% missed scholarship or internship posters in batch chats, while final-year students reported that unmonitored advisor unavailability frequently delays thesis revisions.

4. Proposed Solution and Core User Flow. AcademiaFlow is a semester-adaptive web companion using one-click PDF schedule import: (1) Smart KRS & Circle Sync: students upload their KRS/KSM PDF to preview weekly topics, practicum schedules, and lecturer teaching styles while simulating conflict-free parallel class combinations with their friend circle; (2) Accountable Groupwork Engine: lecturers or students form groups that auto-intersect members' free hours, assign sub-tasks with instant relay notifications alerting the next member upon task completion, stream real-time contribution metrics to lecturers, and offer an objective escalation channel for inactive peers; (3) Adaptive Opportunity & Alumni Hub (Semesters 5–7): the dashboard surfaces filtered scholarships, MoU-verified internships with instant SKS conversion simulators, activity-point (TAK/SKKM) trackers, and opt-in alumni contacts (WA/IG/LinkedIn) at partner companies; and (4) Thesis SLA & Smart Rebalancing: a revision logbook flags 21-day advisor inactivity and recommends replacement advisors to the coordinator by topic match and quota.

5. How It Differs from Existing Alternatives. Legacy portals (SIAKAD, iGracias, SIX) only record registration and final grades, whereas generic chat and kanban tools lack KRS schedule intersection, lecturer contribution telemetry, MoU SKS mapping, and advisor SLA rebalancing. AcademiaFlow unifies these four workflows without requiring write-access to university databases.

6. Planned Technologies. A Next.js 15 and TypeScript web app backed by Supabase (PostgreSQL, pgvector, Auth, and Realtime WebSockets). Collaborative workspaces use BlockNote and Yjs, while a Python FastAPI service uses MarkItDown, PaddleOCR, and Google LangExtract (Gemini API) to parse KRS PDFs, syllabi, and posters into JSON.

7. Expected Benefit and Development Plan. Expected benefit: reduce group coordination overhead and missed deadlines by 60%, curb free-riding in pilot classes, and cut SKS consultation time under 5 minutes in a two-week trial with 30 students. Plan: Week 1 KRS parser, Circle Sync, and Groupwork Engine; Week 2 Opportunity/Alumni Hub and Thesis SLA Rebalancer; Week 3 pilot testing. Unavailable campus APIs are mitigated via PDF parsing.

Source Acknowledgments. [1] Preliminary multi-campus student workflow audit and undergraduate survey on KRS, group coursework, internships, and thesis supervision (November 2026).